

IT481

Smart Refrigerator System

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Abstract

The global challenge of household food waste represents a critical sustainability issue, with homes accounting for 60% of consumer-level waste approximately 79 kilograms per person annually. This waste contributes to economic losses exceeding \$1 trillion globally, generates 8-10% of greenhouse gas emissions, and squanders vast agricultural resources while millions face food insecurity. Current commercial smart refrigerators fail to address the root causes of food waste due to heavy reliance on manual inventory management and lack of automated expiration tracking.

This paper presents an integrated IoT-based Smart Refrigerator System that automates food inventory management, predicts expiration timelines, and reduces waste through intelligent sensing and decision-making. The proposed system adopts a multilayer architecture that includes a sensing layer for capturing item presence, quantity, and freshness conditions; a processing layer for performing local analysis and expiration prediction; and a user interaction layer delivering real-time inventory insights, alerts, and personalized recommendations. The system represents a problem-first approach to smart kitchen technology, offering an accessible, modular, and cost-effective alternative while directly targeting the behavioral and informational gaps that drive household food waste.

1 Introduction

The modern global food system is plagued by a significant paradox, while hundreds of millions of people face hunger, an estimated 1.05 billion tons of food are wasted annually [1]. This issue presents a critical challenge to global sustainability, with profound economic, environmental, and social consequences. This research focuses specifically on the problem of household food waste, which has been identified as the single largest contributor to this crisis, accounting for 60% of all food wasted at the consumer level [2]. On average, this amounts to 79 kilograms of food wasted per person at home each year.

The importance of this problem cannot be overstated. Economically, food waste costs the global economy over \$1 trillion annually [3]. Environmentally, food waste is a major driver of climate change, responsible for 8-10% of all global greenhouse gas emissions [4]. Socially, this waste represents a moral failure, as over one billion meals are thrown away daily while up to 783 million people face chronic hunger [5].

The current state of the art in technological interventions is represented by commercial "smart refrigerators" from major appliance manufacturers like Samsung, LG, Bosch, and others [6]. These devices aim to assist consumers by incorporating features such as Wi-Fi connectivity, large touchscreens, and internal cameras that allow users to view their fridge's contents remotely via a smartphone app [7]. The central promise of these technologies is to provide a connected hub for the kitchen that simplifies food management and helps reduce waste.

However, a significant gap exists between the promise of these commercial systems and their practical efficacy. Current smart refrigerators fail to address the root causes of food waste because they do not automate the most critical information management tasks. Crucially, both inventory logging and expiration date tracking remain overwhelmingly manual processes, requiring the user to tediously input every item and its expiry date, causing cognitive overload.

Widespread adoption of current smart refrigerators is hindered by high costs, repair complexity, and software obsolescence. This research bridges this gap with an IoT framework designed to automate waste reduction. Unlike existing products, this system utilizes multi-modal scanning combining barcode recognition for packaged goods and computer vision with Time-of-Flight (ToF) sensing for fresh produce. Key features include ML-based expiration monitoring, intelligent recipe generation via external APIs, and automated grocery management. By leveraging sensor fusion (weight, ToF, RGB camera, etc.) to eliminate manual data entry, the proposed framework offers a closed-loop, user-centric solution to effectively mitigate household food waste.

This document is organized into five main sections. Section 1 introduces the problem of household food waste and the limitations of existing smart refrigerator solutions. Section 2 presents a literature review of existing research and provides a competitive analysis of current market solutions. Section 3 introduces the proposed Smart Refrigerator System. Section 4 discusses the system's capabilities and potential impact followed by the conclusion, and finally Section 5 presents the future work directions.

2 Literature Review

In this section we will present the research that has been conducted in smart food management systems and the competitors in the smart refrigerator market.

Research into smart food management and refrigerator monitoring systems has gained momentum in recent years due to increasing concerns over food waste, sustainability, and eating healthy habits [8]. A variety of approaches have been explored, ranging from RFID-based embedded systems to IoT sensor networks and advanced computer vision models [9]. These studies highlight different strategies for tracking inventory, predicting spoilage, and improving user interaction, each with its own balance of accuracy, scalability, and practicality.

Parada et al. (2019) [9] developed an RFID-based embedded system for sustainable food management within an IoT framework. The system employs RFID readers connected to an Arduino Mega controller to automatically identify food products, track expiration dates, and recommend recipes based on ingredient availability and user allergies. Each product is equipped with a passive RFID tag containing nutritional and allergen information, which must be manually written during setup. While this enables detailed product identification, it limits scalability when managing a large and frequently changing variety of food items. To

generate recipe recommendations, the authors introduced two algorithms, to make sure that a recipe is only recommended if the combined weight of available key ingredients exceeds that of the missing ones. This strategy allows the system to make realistic suggestions. Despite the fact that this study is not conducted on a refrigerator, it demonstrates how integrating RFID sensing and intelligent recommendation logic can support sustainable food use and reduce household waste.

Velasco et al. (2020) [10] developed an Internet of Things-based refrigerator monitoring system that emphasizes affordability using Arduino microcontrollers and a multi-sensor network. The design integrates weight sensors for freezer and vegetable compartments, ultrasonic sensors for eggs, push button switches for bottled drinks, and a camera for capturing images of the interior. Data from the sensors are processed by an Arduino Uno and transmitted to an Arduino Yun. An Android application allows users to remotely monitor their fridge contents in real time. The system successfully demonstrated compartment-level tracking and wireless accessibility, providing a practical alternative to costly commercial smart fridges. However, its functionality is limited to stock presence detection without predictive intelligence, while challenges such as wiring complexity, camera coverage, and dependence on sensor reliability remain.

Dai (2024) [11] developed a deep-learning approach for robust food recognition in refrigerator environments, focusing on challenges such as occlusion, overlapping objects, and poor lighting conditions. The proposed system integrates a BroadFPN-YOLACT architecture. Evaluated on combined public refrigerator datasets and real fridge images, the system achieved mean average precision of around 48–50 percent. The work highlights the importance of balancing efficiency and reliability in vision-based food computing, while noting limitations in class coverage and performance for unfamiliar food items.

Nong Thuc et al. (2025) [12] presented a smart refrigerator system that integrates IoT sensors and deep learning to support real-time food monitoring. The design uses an ESP32-CAM to capture fridge images every minute, with a YOLO-based model detecting and classifying fruits and vegetables. Sensor data for temperature and humidity were collected alongside image detections. The system achieved a mean average precision of 0.769 at IoU 0.5, demonstrating effective recognition of some items, though performance was weaker for other types of foods. While limited to produce items and not designed for spoilage prediction, the system highlights how computer vision methods combined with low-cost IoT hardware can enhance the robustness of fridge food recognition and support household waste reduction. Compared with Dai (2024) [11], who achieved mAP results around 48–50 percent, Nong Thuc et al. achieved higher detection performance on their dataset but at the cost of narrower scope.

Taken together, these studies show a clear progression in food management research: from early RFID solutions that emphasized data storage and rule-based recommendations, to sensor-driven IoT systems that expanded accessibility, and finally to deep-learning models that enable real-time recognition and monitoring. While each approach addresses important

challenges, limitations remain in terms of scalability, coverage of diverse food categories, and predictive intelligence.

The table below shows a summary of the studies:

Study / Year	Technology Used	Core Functionality	Intelligence Level	APP/Website?
Parada et al. (2019) [9]	RFID + Arduino Mega (IoT-based embedded system)	Food identification, expiry tracking, recipe suggestions	Rule-based algorithm (no machine learning)	None
Velasco et al. (2020) [10]	IoT sensors + Arduino + Dropbox/Temboo + Android app	Real-time fridge monitoring via app	No AI (sensor-based presence detection)	Android application
Dai (2024) [11]	Deep Learning (BroadFPN-YOLACT)	Food recognition, segmentation under occlusion	Deep learning (CNN-based real-time segmentation)	None
Nong Thuc et al. (2025) [12]	IoT + Deep Learning (YOLO-based)	Real-time fruit & vegetable detection	Deep learning (YOLO-based detection)	Web application

Table 1- Summary of studies

2.1 Competitors analysis

In this section, we present four key competitors in the smart refrigerator market, summarizing their purpose, strengths, and limitations to highlight the distinct contributions of our proposed system.

The current smart refrigerator market is defined by advanced connectivity and convenience, yet it remains hindered by a reliance on manual user input. LG ThinQ® [13] and Bosch Home Connect [14] excel in appliance performance and remote diagnostics optimizing energy and climate via their respective apps but both lack automated inventory tracking, forcing users to manually log items. Haier [15] integrates AI for temperature adjustments and freshness monitoring through the hOn App, yet similarly suffers from a lack of automated entry and restocking features. Samsung's Family Hub™ [16], the most visually advanced option, utilizes internal cameras for product recognition and meal planning; however, even this sophisticated system cannot read barcodes for precise expiration dates nor does it employ sensors to predict the actual shelf-life of fresh produce, leaving significant gaps in data accuracy.

Category	Feature	LG ThinQ®	Bosch Home Connect	Haier Smart Refrigerator	Proposed Smart Refrigerator System
Scanning & Recognition	Barcode scanning for regular products	✗	✗	✗	✓
	Camera scanning for fresh products (fruits & veggies)	✗	✗	✓	✓
	Automatic expiration date detection (barcoded items)	✗	✗	✗	✓

<i>Prediction & Monitoring</i>	Predicted expiration (via temp & moisture sensors)	✖	✖	✖	✓
	Expiration alerts for expired items	✖	✓	✓	✓
<i>Smart Food Management</i>	Recipe suggestions before expiry	✖	✓	✓	✓
<i>Inventory Control</i>	Monitor products in/out (TOF sensors)	✖	✖	✖	✓
	Real-time inventory tracking & listing	✖	✓	✓	✓
	List finished/out-of-stock items	✖	✖	✖	✓
<i>Grocery Automation</i>	Automatic grocery tracking & reorder suggestions	✖	✖	✖	✓

Table 2 - Competitors comparison

Consequently, a critical void exists for a fully autonomous food management system. While existing solutions offer robust remote control and climate management, they fail to close the loop on inventory tracking. The market lacks a comprehensive solution that combines barcode scanning for packaged goods accuracy, movement sensors for real-time stock updates, and environmental analytics for true spoilage prediction. Current devices are "smart" at the appliance level (cooling, energy, diagnostics) but remain "passive" at the content level, requiring constant human intervention to maintain a valid digital inventory .

In contrast, the proposed Smart Refrigerator System advances beyond these limitations by combining dual scanning technology barcode-based recognition for packaged goods and AI camera analysis for fresh produce with Time-of-Flight (TOF) sensors to monitor product movement in and out of the fridge. This multi-sensor approach enables precise item detection, predictive expiration modeling through temperature and humidity data, and proactive recipe recommendations to minimize waste. Furthermore, the system introduces automated grocery management and expiration alerts through an integrated mobile platform, providing a comprehensive, intelligent solution that bridges the gaps left by existing smart refrigerator technologies.

3 Proposed System

In this section, we present the complete design and operation of the proposed Smart Refrigerator System. Section 3.1 provides an overview of the system's purpose and overall workflow. Section 3.2 details the hardware architecture, including sensing components, processing units, and communication modules. Section 3.3 explains the sensor placement inside the refrigerator and the data flow between system layers. Section 3.4 outlines the

software architecture used for data processing, AI inference, and cloud integration. Finally, Section 3.5 describes the implementation of key system features such as item recognition, inventory tracking, expiration prediction, and recipe recommendation.

3.1 System Overview

The proposed Smart Refrigerator System is an IoT-based food management platform designed to automatically monitor food inventory, predict expiration dates, and suggest recipes to reduce waste and promote healthy consumption. The system operates through three main layers: sensing, processing, and user interaction.

In the sensing layer, a combination of Time-of-Flight sensors, RGB cameras, barcode scanners, weight, and environmental sensors capture visual, spatial, and contextual data from inside the refrigerator. A door sensor triggers data collection whenever the refrigerator is accessed.

The processing layer employs a Jetson Nano as the main computing unit and an ESP32 for sensor integration. Data from multiple sources are fused and analyzed using YOLOv8-based object detection [17] and an expiration-prediction model, with results stored in a cloud-connected database for real-time tracking.

Finally, the user interaction layer is implemented through a mobile application developed using Flutter, which displays inventory status, expiration alerts, and recipe recommendations. This ensures user convenience and seamless access to real-time information without requiring a built-in refrigerator display.

Figure 1 shows a graph demonstrating the system architecture overview.

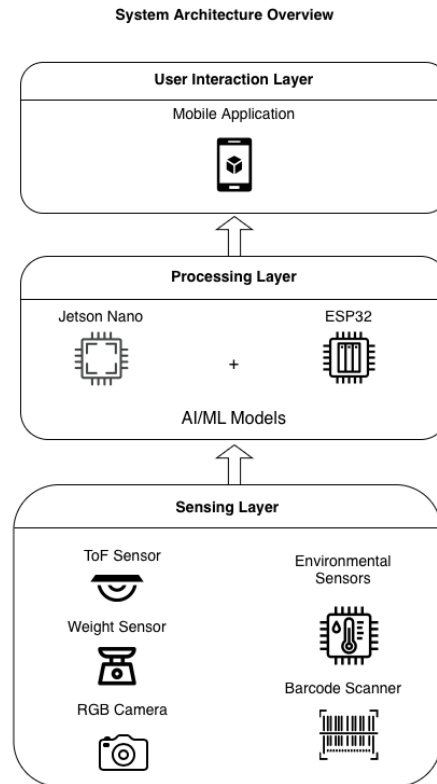


Figure 1 System Overview

3.2 Hardware Architecture

This section describes the physical components that enable sensing, processing, and communication within the Smart Refrigerator System.

3.2.1 Sensing Components

A. ToF sensor (Time-of-Flight sensor)

The ToF sensor is used to detect items in the refrigerator even when they are partially hidden or stacked behind others, it produces 3D depth maps that enable the detection of stacked or occluded items.

B. RGB camera

High-resolution RGB cameras such as 5 MP USB cameras capture color images of the refrigerator's interior for visual identification of unpackaged items. The captured frames are processed by deep learning models such as YOLOv8 [17] to classify fruits, vegetables, and other untaged products.

C. Barcode scanner (packaged item recognition)

A compact laser barcode scanner (e.g., Zebra DS457) is installed near the refrigerator door to scan packaged products, allowing automatic retrieval of product names and expiration details from the database.

D. Environmental sensors (temperature, humidity, ethylene gas)

The system includes temperature sensors, humidity sensors, and an ethylene gas sensor. These sensors continuously measure internal environmental parameters.

E. Weight sensors

Each shelf is equipped with a load cell capable of detecting small variations in weight with a sensitivity of a few grams. This enables precise detection of item placement, removal, and quantity estimation for inventory tracking.

F. Door sensor (system activation trigger)

A door sensor is attached to the refrigerator frame to detect door opening and closing events. When the door is opened, the sensor triggers the activation of the data acquisition and image-capture process.

3.2.2 Processing & Communication Unit

A. Jetson Nano

The Jetson Nano [18] serves as the main edge computing unit responsible for AI computation, data fusion, and local database operations.

B. ESP32 Microcontroller

The ESP32 [19] functions as the sensor control and communication unit.

C. Wi-Fi module and cloud connectivity

The built-in Wi-Fi capability of the ESP32 allows seamless connection between the refrigerator and cloud services.

D. Local storage and power supply

A microSD card is used for local data storage, including captured images, sensor readings, and backup files. Energy efficiency is maintained through power management features such as sleep mode activation when the refrigerator door is closed.

3.2.3 User Interface Devices

The user interface of the Smart Refrigerator System is implemented entirely through a mobile application, which serves as the primary platform for interaction. The mobile app, developed using Flutter, provides real-time access to inventory information, expiration alerts, environmental conditions, creating shopping lists, and recipe recommendations.

3.3 Sensor Placement and Data Flow

This section describes how sensors are strategically positioned inside the refrigerator to capture depth, visual, environmental, and weight data. It also explains the flow of information from the sensing layer to the processing unit and finally to the mobile application for real-time monitoring.

A) Refrigerator Layout

The sensors are distributed across the refrigerator to ensure full coverage of all storage areas. Time-of-Flight sensors are mounted at the top of each shelf to generate depth maps and detect stacked or hidden items. RGB cameras are positioned to capture clear visual images of the main compartments. Weight sensors are placed beneath each shelf to detect the

item placement or removal, while temperature, humidity, and ethylene gas sensors are installed in compartments such as produce drawers where freshness monitoring is essential. A barcode scanner is positioned near the door area to capture packaged items as they are placed inside. The door sensor is mounted on the refrigerator frame to initiate data collection upon access.

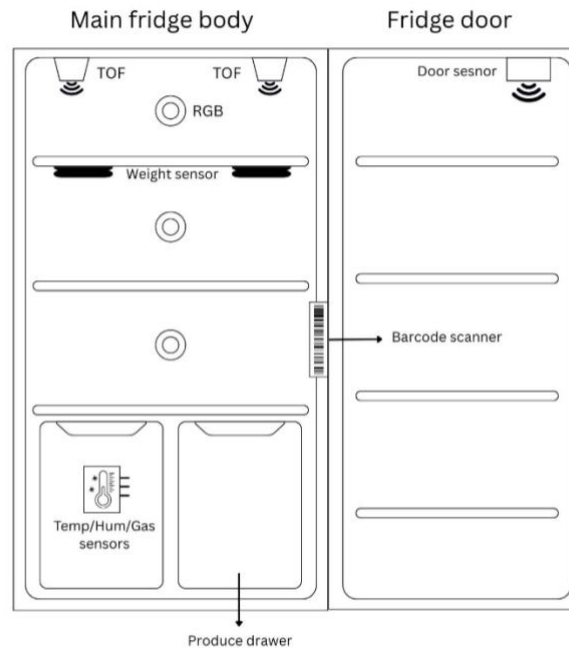


Figure 2 Refrigerator Layout

B) Sensor Coordination and Trigger Mechanisms

Sensor activity within the refrigerator is coordinated to optimize energy use and ensure accurate data capture. The door sensor acts as the primary trigger, activating ToF depth mapping and camera capture only when the refrigerator is open. Weight sensors detect load changes when removing or adding items. Environmental sensors gather readings periodically to monitor freshness conditions without requiring door activity. This coordinated sensing strategy reduces unnecessary processing while ensuring high-quality data for item detection and inventory updates.

C) Data Flow

These raw sensor readings are transmitted to the ESP32, which organizes the data into structured packets for processing. The ESP32 then sends the collected data to the Jetson Nano. The Jetson Nano first performs data fusion to combine depth, visual, weight, and environmental inputs into a single coherent representation of the refrigerator's contents. Following this fusion step, AI/ML models are applied to detect and classify items, identify their location on specific shelves, and evaluate freshness conditions. The expiration-prediction model further estimates remaining shelf life.

After processing, the Jetson Nano formats the results into inventory records and transmits them to the cloud database. The mobile application retrieves this information in real time, ensuring that users always have access to the latest inventory status, freshness alerts, and recipe suggestions based on available ingredients.

3.4 Software Architecture

This section describes the software components that enable data processing, intelligent analysis, and user interaction within the Smart Refrigerator System.

3.4.1 Embedded Software

A. Operating System and Programming Environment

The Jetson Nano provides a stable Linux environment for running computational tasks. Python 3.9 serves as the primary programming language. The ESP32 microcontroller is programmed using the Arduino framework (C/C++).

B. Computer Vision Framework (OpenCV)

OpenCV (Open-Source Computer Vision Library) is used for image processing and manipulation tasks. This framework handles operations such as image capture from RGB cameras, preprocessing for quality enhancement, and frame extraction for analysis.

C. Object Detection Model (YOLOv8)

YOLOv8 (You Only Look Once version 8) [17] is a deep learning model responsible for detecting and classifying food items captured by the RGB cameras. YOLOv8 is optimized for edge deployment using TensorFlow Lite, which uses of the Jetson Nano's GPU acceleration capabilities to efficiently execute on limited hardware resources.

D. Depth Processing Module

A depth processing module analyzes data from the ToF sensors to generate three-dimensional depth maps of the refrigerator's interior. The output is a spatial representation of item locations across different shelves.

E. Expiration Prediction Model

The expiration prediction model is a machine learning component that estimates the remaining shelf life of fresh produce. It receives inputs from environmental sensors and product metadata, then predicted the expiration date or days remaining until spoilage. For packaged items, the model retrieves expiration dates from barcode databases.

F. Local Database (SQLite)

A local SQLite database is embedded within the Jetson Nano to store real-time inventory data, sensor readings, and processing results.

G. Local API Service (Flask)

A lightweight RESTful API service is implemented using Flask to facilitate communication between system components. This API serves as the internal communication interface, allowing the ESP32 to send sensor data to the Jetson Nano and enabling the Jetson Nano to transmit processed results to the cloud backend.

3.4.2 Cloud Components

The cloud backend layer serves as the central data repository and coordination hub, ensuring synchronization between the edge devices and the mobile application.

A. Backend Server Framework (Node.js + Express)

The backend server is built using Node.js with the Express framework. This server handles incoming requests from both the Jetson Nano and the mobile application, processes API calls, and manages data flow between system components. The server is hosted on AWS using AWS EC2 instances.

B. Cloud Database (Firebase Realtime Database)

Firebase Realtime Database is used to store all persistent data for the system, including inventory records, user profiles, historical sensor data. The database stores data in a JSON tree structure and is accessed by both the backend server and mobile application for reading and writing information.

C. IoT Communication Service (AWS IoT Core)

AWS IoT Core provides secure, scalable connectivity between the Jetson Nano and the cloud infrastructure. This service uses MQTT (Message Queuing Telemetry Transport) protocols to enable publish-subscribe messaging, allowing the Jetson Nano to send inventory updates and the mobile app to receive real-time notifications efficiently.

D. External Recipe API (Spoonacular)

The system integrates with the Spoonacular API to fetch recipe suggestions based on available ingredients. This API is queried by the backend server with a list of items currently in the refrigerator, and it returns recipes that match the available ingredients.

E. Barcode Lookup API (Open Food Facts)

The Open Food Facts API is used to retrieve product information when a barcode is scanned. The retrieved data is then stored in the Firebase database and displayed to the user through the mobile application.

3.4.3 Mobile Application Components

The mobile application serves as the primary user interface, providing access to inventory information, alerts, and system features.

A. Mobile Framework (Flutter)

The mobile application is developed using Flutter, a cross-platform framework developed by Google that allows a single codebase to be deployed on both iOS and Android devices.

B. Authentication Service (Firebase Authentication)

Firebase Authentication handles user account creation, login, and session management using email/password or third-party providers like Google Sign-In. It integrates with Firebase Realtime Database to enforce secure access, ensuring each user can only view their own refrigerator data.

C. Real-Time Synchronization Service (Firebase Listeners)

Firebase Realtime Database listeners keep the mobile app continuously synced with the cloud. Any inventory update is pushed to the app instantly without manual refresh, ensuring always-current information.

D. Local Storage (Hive)

The mobile app includes local storage using Hive, a lightweight and fast NoSQL database designed specifically for Flutter applications. Hive caches inventory data and user settings locally on the device, allowing the app to display the most recent inventory state.

3.5 Feature Implementation

3.5.1 Product Scanning and Recognition

The product scanning and recognition feature enables automatic identification of food items through two complementary approaches: barcode scanning for packaged products and camera-based object detection for unpackaged items.

A. Barcode Scanning for Packaged Items

When the door sensor detects refrigerator access, the barcode scanning system activates with a dual-scanner approach. A dedicated laser barcode scanner positioned near the door entrance operates alongside an RGB camera, both working simultaneously to ensure comprehensive barcode detection.

As packaged products pass through the entrance zone, the Zebra scanner continuously scans for barcodes using its focused laser beam. Simultaneously, the RGB camera captures continuous frames of the entrance area. OpenCV algorithms process these frames to detect barcode patterns including UPC, EAN-13, and QR codes. This dual-scanner configuration ensures that barcodes are detected regardless of item orientation or positioning -the laser scanner captures barcodes aligned with its beam, while the camera-based system detects barcodes at various angles and positions. Once detected by either scanner, the decoding algorithm extracts the unique product identifier.

The system supports multiple scanning methods to accommodate different user preferences and situations, users can also scan barcodes using their smartphone through the mobile application, where the app activates the phone's camera and decodes the barcode directly. The decoded barcode is transmitted to the Open Food Facts API, which returns comprehensive product information including name, brand, category, and nutritional data within 1-2 seconds.

B. Camera and ToF Detection for Unpackaged Items

For unpackaged items such as fruits, vegetables, and prepared foods, the system combines RGB camera-based object detection with ToF depth sensing. The RGB camera captures high-resolution images that are processed by the YOLOv8 model, which classifies objects into specific categories with confidence scores. Simultaneously, ToF sensors generate depth maps, enabling detection of items that are partially hidden or stacked.

For containers holding prepared foods, the system recognizes common container types through visual classification but prompts users via the mobile app to identify contents, which

are then associated with the detected container location. Environmental sensor data is also associated with the item to enable more accurate expiration predictions.

3.5.2 3D Depth Mapping

ToF sensors mounted at the top of each shelf continuously emit infrared light pulses, measuring the time taken for reflection to calculate distance to each surface point. This process generates depth maps, allowing real-time tracking of object placement and movement. When items are stacked or overlapping, a standard 2D camera can only detect topmost visible items, but the depth map reveals the three-dimensional structure through distinct height clusters at different distances from the sensor. Clustering algorithms segment these depth regions into separate objects, which are then matched with visual data from the RGB camera for classification. This fusion aligns spatial coordinates from both sensors, ensuring each detected object has both visual appearance information and precise spatial location.

3.5.3 Expiration Prediction

The system employs two distinct approaches for expiration prediction: rule-based prediction for packaged items and machine learning-based prediction for fresh produce.

For barcode-scanned packaged products, if available, the expiration date is used directly for countdown tracking, it's calculated by adding this duration to the current date, with adjustments made based on the refrigerator's average temperature when it consistently exceeds recommended ranges.

Fresh fruits and vegetables require a different approach since they lack printed expiration dates and their shelf life varies significantly based on environmental conditions. The system employs a TensorFlow Lite regression neural network that predicts remaining shelf life using multiple inputs including product type, current temperature, humidity level, ethylene gas concentration, initial visual freshness assessment extracted from camera images, and time elapsed since addition. The model outputs predicted days until spoilage, and this prediction can be refined periodically with updated sensor data to account for changes in storage conditions.

Once expiration dates are assigned, the system continuously monitors the countdown for all items through a daily background process. Items are classified into three categories based on remaining time: Fresh with more than 3 days remaining, Expiring Soon with 1-3 days remaining, and expired with zero or negative days remaining. Items in the Expiring Soon category trigger proactive notifications through the mobile app and activate the recipe suggestion feature to recommend dishes utilizing these ingredients. Expired items generate urgent alerts and are flagged in red within the inventory interface to prompt immediate action.

3.5.4 Inventory Tracking

The inventory tracking system monitors the entry and exits of items in real time through multi-sensor data fusion, maintaining an accurate record of what is currently stored in the refrigerator.

When the door sensor detects refrigerator access, the system enters active monitoring state. Weight sensors beneath each shelf record baseline values and detect load changes as items are added or removed. Simultaneously, ToF sensors and cameras capture the shelf state before and after each interaction. For item entry, the system identifies new objects by comparing depth maps and correlates weight increases with detected objects by matching timestamps and shelf locations. Once successfully identified, a new inventory record is created in the database with all relevant details including item name, category, weight, shelf location, and predicted expiration date.

For item removal, weight sensors detect load decreases while ToF sensors capture updated depth maps showing object disappearance. The removed item's inventory record is updated to mark it as consumed and moved to historical logs. If the item was the last unit of its type, the system automatically adds that product to the grocery shopping list. Users can review this auto-generated list and manually add or remove items as needed.

3.5.5 Recipe Suggestions

The recipe suggestion feature integrates with the Spoonacular Recipe API to provide cooking recommendations based on current inventory, particularly prioritizing ingredients approaching expiration. The system is activated when one or more items reach the "Expiring Soon" status (1-3 days remaining).

When recipe suggestions are needed, the backend server constructs an API query containing the list of available ingredients extracted from the current inventory. The API returns a list of recipes that can be prepared using some or all the specified ingredients.

The raw recipe results are filtered and ranked based on several criteria to ensure relevance and practicality. First, the system identifies the main ingredient of each recipe, typically the primary protein or base component. Recipes whose main ingredients are not available are excluded from the results. Next, recipes that use expiring ingredients are prioritized and moved to the top of the list to help reduce food waste. Finally, recipes that utilize the maximum number of available ingredients are ranked higher to minimize the need for additional purchases. The filtered and ranked recipe list is then sent to the mobile application for display.

4 Conclusion and discussion

This work set out to address the persistent global challenge of household food waste by designing a smart, IoT-enabled refrigerator system capable of the features mentioned throughout this paper. The proposed system successfully demonstrates how an integrated

sensing, processing, and user-interaction framework can transform traditional refrigerators into intelligent food management hubs that operate with minimal user burden.

The discussion of our findings highlights several important contributions. First, the system proved that automated inventory logging is achievable with high reliability when combining different technologies. This multi-modal sensing strategy overcame many limitations found in current commercial products, where camera-only systems fail to detect hidden or stacked items and manual input remains the dominant inventory method.

Second, the integration of machine learning-based expiration prediction and environmental monitoring enabled the system to go beyond simple item tracking. Outperforming static date-based methods found in most current smart appliances. This predictive capability is essential for addressing non-packaged foods, which represent a significant portion of household waste and are often overlooked by commercial solutions.

Third, the mobile application played a critical role in translating system intelligence into meaningful user engagement. This supports the broader goal of reducing cognitive load, a major factor in household waste behaviors. The system's reliance on a mobile interface rather than a built-in refrigerator display also demonstrated advantages in cost, scalability, and long-term maintainability.

The proposed architecture not only improves operational efficiency but also enhances accessibility and affordability compared with high-end commercial smart refrigerators. By relying on modular IoT components, edge computing, and cloud-based synchronization, the solution remains adaptable and upgradable without requiring full appliance replacement.

5 Future Work

Several opportunities exist to further expand and refine the system. One direction is the integration of more advanced on-device AI models capable of classification of diverse food categories, including leftovers and cooked meals. Future enhancements may also include automated container recognition, improved occlusion handling using multi-camera arrays, or integration with retail APIs for grocery delivery automation. On a larger scale, the system's architecture can be extended to contribute to global sustainability goals by supporting large-scale data analytics on food consumption and waste patterns.

References

- [1] F. Harvey, "Millions go hungry as a billion meals binned every day, says UN report," The Guardian, 7 March 2024. [Online]. Available: <https://www.theguardian.com/environment/2024/mar/27/fifth-of-food-wasted-globally-at-cost-of-1tn-a-year-says-un-report>. [Accessed 1 November 2015].
- [2] BioCycle, "UN's 2024 Food Waste Index Attributes 60% Of Food Waste To Households," BioCycle, 9 April 2024. [Online]. Available: <https://www.biocycle.net/2024-food-waste-index/>. [Accessed 1 November 2025].
- [3] Food Manufacturing, "Food Waste Costs the World's Economy \$1T Each Year," Food Manufacturing, 26 June 2023. [Online]. Available: <https://www.foodmanufacturing.com/supply-chain/blog/22865828/food-waste-costs-the-worlds-economy-1t-each-year>. [Accessed 1 November 2025].
- [4] Lactic, "Impact of Food Waste on Environment and Economy," Lactic, [Online]. Available: <https://www.lactic.com/news/impact-food-waste-environment-and-economy>. [Accessed 1 November 2025].
- [5] United Nations, "International Day of Awareness on Food Loss and Waste Reduction," United Nations, [Online]. Available: <https://www.un.org/en/observances/end-food-waste-day>. [Accessed 1 November 2025].
- [6] D. a. M. K. Johnson, "The 7 Best Smart Refrigerators, Vetted By Experts," Forbes, 20 May 2024. [Online]. Available: <https://www.forbes.com/sites/forbes-personal-shopper/article/best-smart-refrigerators/>. [Accessed 1 November 2025].
- [7] SlashGear, "Every Major Smart Fridge Brand Ranked Worst To Best," SlashGear, [Online]. Available: <https://www.slashgear.com/1761812/major-smart-fridge-brands-ranked-worst-best/>. [Accessed 1 November 2025].
- [8] S. Ahmadzadeh, T. Ajmal, R. Ramanathan and Y. Duan, "A Comprehensive Review on Food Waste Reduction Based on IoT and Big Data Technologies," *Sustainability*, vol. 15, no. 4, p. 3482, 2023.
- [9] R. Parada, A. Palazón, C. Monzón López and J. Melià-Seguí, "RFID Based Embedded System for Sustainable Food Management in an IoT Network Paradigm," *Future Internet*, vol. 11, no. 9, p. 189, 2019.
- [10] J. Velasco, L. Alberto, H. D. Ambatali, M. Canilang, V. Daria, J. B. Liwanag and G. A. Madrigal, "Internet of things-based (IoT) inventory monitoring refrigerator using Arduino sensor network," *Indonesian Journal of Electrical Engineering and Computer Science*, vol. 18, no. 1, p. 508–515, 2020.

- [11] X. Dai, "Robust deep-learning based refrigerator food recognition," *Frontiers in Artificial Intelligence*, vol. 7, p. Art. 1442948., 2024.
- [12] K. N. Thuc, K. T. N. Anh, T. N. Huy, D. N. H. Hong and K. D. Ba, "A smart fridge with AI-enabled food computing," *arXiv preprint*, 2025.
- [13] LG, "LG ThinQ," LG, [Online]. Available: <https://www.lg.com/us/lg-thinq>. [Accessed 10 November 2025].
- [14] Bosch, "Smart refrigeration | Home Connect," Bosch, [Online]. Available: <https://www.bosch-home.com/mt/bosch-innovations/home-connect/smart-refrigeration#anc-15797555>. [Accessed 10 November 2025].
- [15] hOn, "hOn Smart functions," hOn, [Online]. Available: <https://hon-smarthome.com/smart-functions/>. [Accessed 10 November 2025].
- [16] Samsung, "Family Hub™ Refrigerators," Samsung, [Online]. Available: <https://www.samsung.com/us/explore/family-hub-refrigerator/overview/>. [Accessed 10 November 2025].
- [17] G. Jocher, A. Chaurasia and J. Qiu, "Ultralytics YOLO," 2023. [Online]. Available: <https://github.com/ultralytics/ultralytics>.
- [18] N. Corporation, "Jetson Nano Developer Kit User Guide," 2021.
- [19] E. Systems, "ESP32 Series: Datasheet," Espressif Systems, 2023.